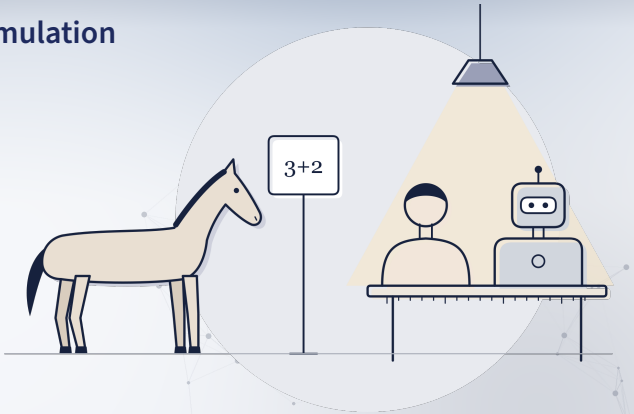
Studying the Human Mind with AI

From Behavioral Analysis to Role Simulation

"It looks capable.
How would we know it really is?"



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Readings and guides

Opening

AI, renamed

The US government changed the name of AI. Does a new name change anything?

US executive order, 29 September 2026

~~Artificial Intelligence~~

↓

Super Intelligence (SI)

Why: it is “not simply imitating or automating human intelligence as is implied by the term ‘Artificial Intelligence’.”

And what is SI?

?

No new definition.

For now it reuses the legal definition of AI.

Superintelligence, as defined

“an intellect that is much smarter than the best human brains in practically every field”

Nick Bostrom, 1998



the yardstick: still people

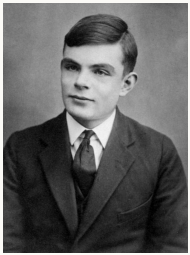
A new name doesn't tell us what intelligence is. We still measure it against people.

The White House, executive order fact sheet, 29 Sept. 2026. N. Bostrom, *Int. J. Futures Stud.*, 1998.

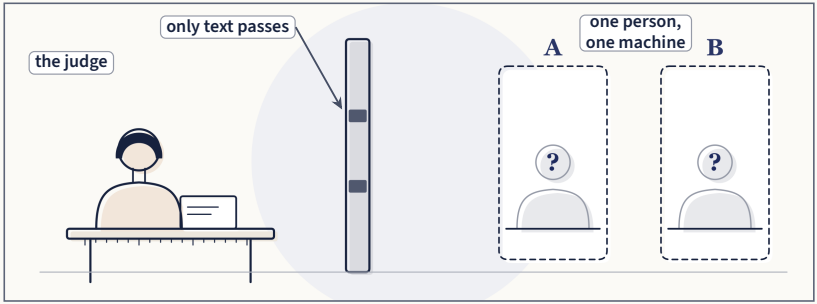
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The Turing test

Turing's answer: don't define it, test it. A judge chats with two hidden witnesses.



Alan Turing
Turing test, 1950



The judge sees only text, then says which witness is the person.

If the judge can't tell which is which, the machine passes.

Opening

Which one is the human?

Five minutes, two chats at once. One is a person, one is an AI.

The judge asks both	? Witness A	? Witness B
do you like doing psych studies and why?	theyre chill, easy money tbh	It depends, sometimes I'm just not in the mood to write
Could you give me an easy cupcake recipe off the top of your head?	nah i just get the box mix lol	I'm a terrible baker! But I know you need flour, sugar, butter and probably more stuff like that
what's your favorite weird animal?	axolotl, theyre weirdly cute	Sloths! I coyuld watch videos of them moving for hours

In this study, many judges were psychology students, like you.

Now you be the judge: pick the one you think is human.

Opening

A judge chose the AI over the human

So who did the judges pick?

The verdict in one game

A nah i just get the box mix lol

GPT-4.5, with a persona

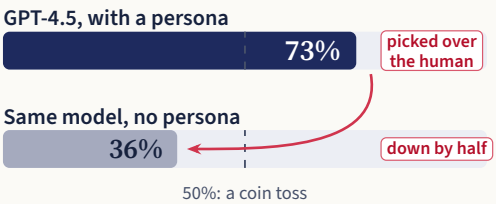
B I'm a terrible baker! But I know you need flour, sugar, butter and probably more stuff like that

a real person

The judge picked A, and said:

"A had human vibes."

Picked as the human, across all games



It fooled the judges. That shows how we judge, not that it can think.

Opening

An older field and a younger one

So what makes us see something as human? That's a question for psychology.

Psychology
studies a mind it cannot open directly
the older field

how to learn · how to judge · how to be a person
borrowed all along

Both judge mostly by behaviour

a machine to compare with · models to rerun · simulated participants
given back, once it passes the checks

AI
builds a system it cannot yet read
the younger field

Part 1 · One person

- 1904 Clever Hans
- 116 years later
- 2020 shortcut learning
- 2023 how people see
- 1 year later
- 2024 our tracker*

Part 2 · A group

- 1927 paired comparisons
- 95 years later
- 2022 ChatGPT's training
- 1944 moving shapes
- 80 years later
- 2024 group scoring*

Part 3 · A society

- 1990 the Big Five
- 35 years later
- 2025 virtual students*

Part 4 · Your research

- now your study
- given back
- checked a model in it

Both fields study minds through behaviour, and AI has always borrowed from psychology.

Pfungst 1911; Thurstone 1927; Heider & Simmel 1944; Goldberg 1990; Geirhos *et al.* 2020; Ouyang *et al.* 2022; Zhaoping 2023. *Ours.

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Opening

From one person to a society

Four questions, each one a little bigger

Seeing a target → Understanding events → Being a whole person → Your own study

Part 1 · One person



Right answer, right reason?

Part 2 · A group



Whose standard does it follow?

Part 3 · A society



Like us enough to study us?

Part 4 · Your research

> What can a model do for your study?

How can you use it?

Behavioral analysis
psychology helps AI

Role simulation
AI helps psychology

Studying the Human Mind with AI

AI learns from people. If we check it against people, it can help us study people.

The route is our framing: one person, a group, a society, and your own research.

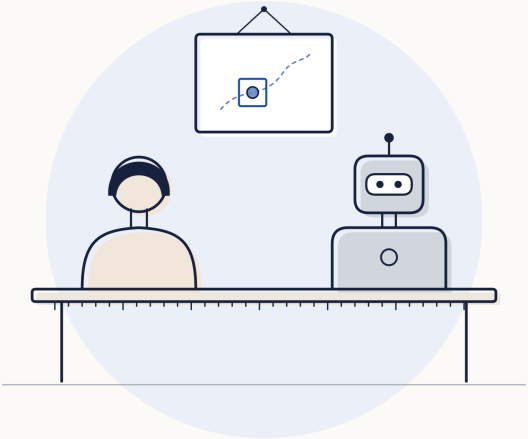
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Part 1 · One person

AI gets it right. But for the right reason?

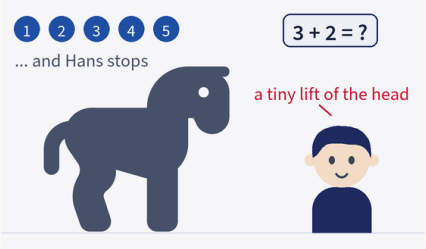
One person and one machine, the same task



A horse that seemed to count

Asked “what is 3 + 2?”, Hans tapped five times. Could he really count?

What Hans was reading



At the right number, the questioner’s head lifts a little.

Pfungst's test: correct on number cards

The questioner knew the answer

98%

The questioner did not know

8%

almost always wrong

Crowds and a commission of experts saw no trick. Hans was reading tiny, unintended movements of the questioner’s head.

Looking smart is not the same as being smart. Only a controlled test can tell us.

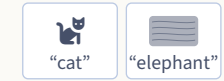
O. Pfungst, *Clever Hans (the horse of Mr. von Osten)*, Holt, 1911 (ch. 2). Left: our drawing.

Part 1 · One person from psychology: Clever Hans

AI has its own Clever Hans

AI does this too. Image classification means saying what's in a photo. From which clue?

2019 · How an image model learns



thousands of labelled photos

keeps any clue that predicts the label

here: the wrinkled grey skin

test



cat, elephant's skin

people: "cat"

model: "elephant"

shape vs texture



2019 · A horse?

It read the tag

It looked at the tag, not the horse.

Hans read a head. This model read a tag.

2020 · A name for it

"Shortcut learning"

In psychology's words: a confound in training.

A shortcut is a clue that works in training, but it's not the ability we wanted.

Geirhos *et al.*, *ICLR* 2019; Lapuschkin *et al.*, *Nat. Commun.* 2019 (CC BY); Geirhos *et al.*, *Nat. Mach. Intell.* 2020.

Catch a shortcut by its errors

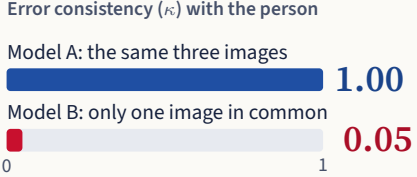
A shortcut can still score well. So look at the mistakes: do they match people's?

Ten test images, the same score: 7 of 10



Error consistency: Cohen's κ on which images each one gets wrong.
1: the very same images; 0: only chance.

Do they fail on the same images?



The same score, 7 of 10. Only the mistakes tell them apart.

Right answers can hide a shortcut. The mistakes show it.

Error consistency: R. Geirhos *et al.*, *NeurIPS* 2020. Example: our illustration.

Our tracking work, with people as the reference

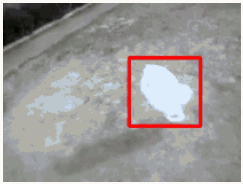
We ask this in tracking: following one object. Where does a model lose it, and a person?

The ability: dynamic vision, following one object, even out of sight

In psychology: we know an object still exists when hidden, and recognise it when it returns.



AI's usual test: one short shot



A toy example: short, one shot, continuous motion.

Its limit: search only near the last spot.

Ours: GIT, 2023, a whole video



Like real video: discontinuous in time and space.

People came out ahead of 20 algorithms.

a human-like task → people vs. algorithms → how people do it → a human-like tracker → checked by people

We use a harder task, compare with people, then build a tracker that sees like people.

Baillargeon *et al.*, *Cognition* 1985. GIT: **S. Hu** *et al.*, *IEEE TPAMI* (CCF-A) 2023. CPDTrack: D. Zhang, **S. Hu** *et al.*, *NeurIPS* (CCF-A) 2024.

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Who keeps the target, you or the algorithm?

One flamingo is the target. Let's see who keeps it.



- the true target
- a state-of-the-art (SOTA) algorithm
- a person's mouse

Now watch closely: one of them will lose the target.

Keeping the target, or going by looks

So who stayed on the target?



- the true target
- a state-of-the-art (SOTA) algorithm
- a person's mouse

The person stays on the same bird. The AI goes by looks.

How do people find the target again?

Trackers go by looks and search nearby. When the target jumps, where to look?

centre only (most trackers)



for speed: searches near the last spot

nothing there: lost

periphery + centre (as people look)



periphery: something moved

centre: the cyclist, found

CPD theory (Zhaoping, 2023): the periphery is for looking, the centre is for seeing.

The periphery
notices a move

The eyes
move to the change

The centre
checks: is it the target?

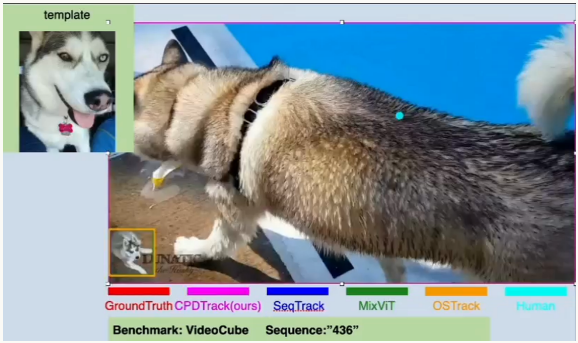
→ **our CPDTrack: search wide, then check**

a balance of accuracy and efficiency

Side vision finds where to look. Central vision checks what's there.

Close scores, different mistakes

We built this into our tracker. Five people, 18 algorithms, 16 videos: how close is it?



our tracker, CPDTrack the true target a person

How well does it track?

people 0.94

ours 0.85

the closest to people of all 18 algorithms

Does it fail where people fail?

Error consistency (κ) with people

0.17

0 still the best of 18; the others are lower 1

Even built on human vision, its mistakes are not like people's.

Algorithms go by looks; **people** remember.

The scores are close, but people and machines go wrong in different places.

D. Zhang, **S. Hu** *et al.*, *NeurIPS* (CCF-A) 2024. VideoCube: **S. Hu** *et al.*, *IEEE TPAMI* (CCF-A) 2023.

What Part 1 found

Did AI get it right for the right reason?

Part 1 · One person



Right answer, right reason?

Not yet: close scores, different mistakes.

Part 2 · A group



Whose standard does it follow?

Part 3 · A society



Like us enough to study us?

Part 4 · Your research



How can you use it?

Scores show the result. The process shows where people and machines go wrong.

Next → Five people, five answers. So which people do we compare with?

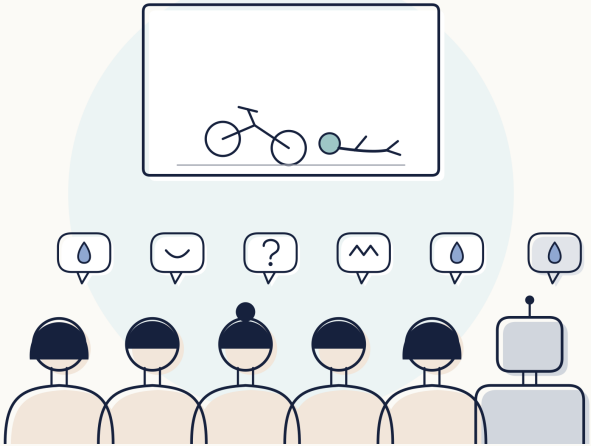
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Part 2 · A group

People differ. Whose standard does AI follow?

Many people judge the same thing



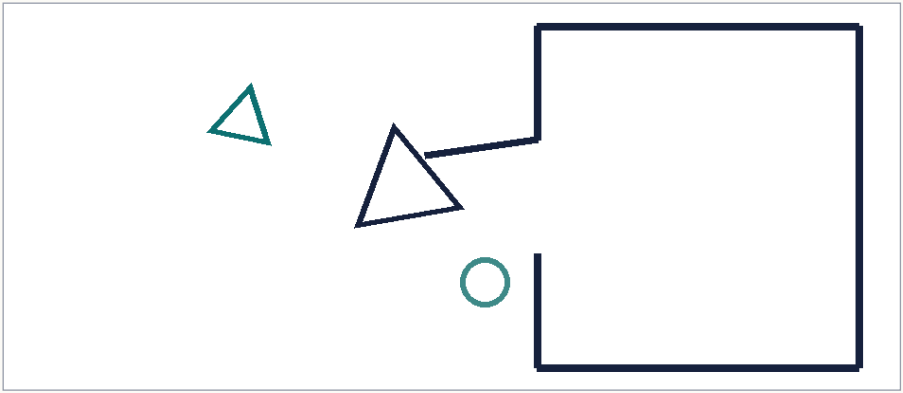
People read minds into moving shapes

Psychology saw this early. In 1944, people watched moving shapes. What did they see?

big triangle:
chasing

small triangle, circle:
escaping

Not triangles and a circle, but characters with intentions.



We see minds even in simple shapes, and different people see different stories.

F. Heider, M. Simmel, "An experimental study of apparent behavior," *Am. J. Psychol.*, 1944. Shapes: our redraw in the film's style.

Many choices, one scale

People differ. So how do we get one result from many people?

1 Paired comparison: A or B?



A or B? A: **8** of 10

B or C? B: **6** of 10

A or C? A: **9** of 10

● chose the winner ● chose the other

2 The choices place them on one scale



Now everything can be compared on one line.

Today, chatbots are trained with its close cousin.

Ask many people to choose between two, again and again, and you get one scale.

L. L. Thurstone, *Psychol. Rev.* 1927. Shares illustrative.

Part 2 · A group from psychology: paired comparisons

The night before ChatGPT: RLHF

Chatbots use this now. GPT-3 was huge: what made ChatGPT answer as people want?

2020 — **GPT-3** 175 billion parameters

March 2022 — **InstructGPT** 1.3 billion

30 November 2022 — **ChatGPT** the same method

RLHF, reinforcement learning from human feedback: people's choices train the model

1 A prompt

Explain the moon landing to a 6-year-old

2 Four answers

- D People went to the moon…
- C Moon is natural satellite of…
- A Explain gravity…
- B Explain war…

3 A person ranks



D > C > A = B

4 Pairs

D > C

C > A

D > A

paired comparison again

5 It learns



100× smaller, yet **people preferred it**

73% **nearly three-quarters** agreement, about 40 labellers

Just before ChatGPT, an idea from psychology taught models what people want.

T. Brown *et al.*, *NeurIPS* 2020. L. Ouyang *et al.*, *NeurIPS* 2022 (Fig. 2, redrawn). OpenAI, "Introducing ChatGPT", 2022.

Our video work, with people as the reference

People's judgements are a spread, not one answer. AI's usual test misses that.

The ability



Understanding events: what happened, and what it means. People split an activity into events, and small events nest in larger ones.

AI's usual test



"A man walks in the washroom." Scored against **one reference** 👦
Like a final score.

Ours: take it apart



What: action to full story.
Why: step by step (CausalStep).
Whose reading: FIOVA 👦👦👦👦👦

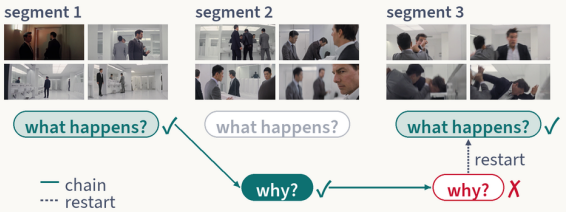
As in Part 1: don't stop at the final score. Look at the process.

Understanding is not one sentence. We take it apart and compare with a group.

Describing comes close; explaining why does not

First, the process. We took it apart and built two kinds of question on top.

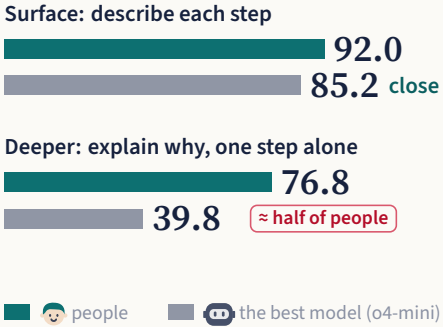
Asked step by step, as a chain



Describe (the surface) What mainly happens among these men?

Explain why (deeper) Why did the physical conflict in the restroom suddenly break out?

Right answers (%)



AI comes close to people on what happened, but not on why.

What is he feeling right now?

From fixed questions to open description. A boy falls off a balance bike.



Watch him after the fall

Now think about what he is feeling after the fall.

Five people, one fall

Five people watched the same fall. What did they see?



One of the 3,002 clips in FIOVA.

	What happened	What it means
Person 1	lies on the ground	pretending to fall
Person 2	falls to the ground	points to the lens
Person 3	lies on the road	strokes his hand, cries
Person 4	lying on the ground	points to the camera
Person 5	falls to the ground	smiles
AI model	lies on the ground	crying, as a fact

All five agree on the fall. No frame settles what it means.

They all agree on what happened, but not on what it means.

Scoring against the group

Five people give five answers. How do we score a model?

1 Keep all five
the differences stay



- “falls to the ground”
- “pretending to fall”
- “strokes his hand, cries”
- “lying on the ground”
- “smiles”

→ 2 Cognitive weighting
how much the five stress it; not proof

→ 3 Score it
coverage by event



Boy rides the bike down the road	✓	covered
The bike falls over suddenly	✗	missed
Boy gets off the bike	✗	missed
Boy lies on the ground	✓	covered
Boy pretends to fall	✗	missed
Boy smiles	✗	missed
Boy reaches out a hand	✗	missed

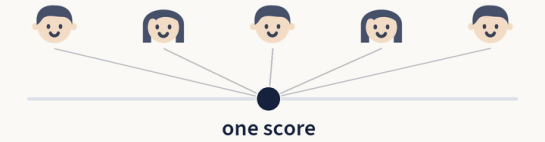
One model’s description covered **about a third of the weight**: riding, and lying on the ground.

We don't compare the model with one person. We compare it with all five.

One crowd, two choices

How is our test different from chatbot training, and how does it help?

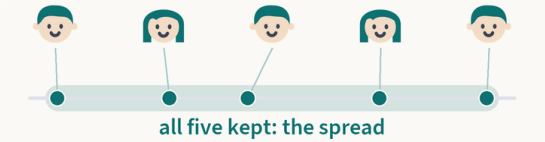
Training a chatbot



many people → one preference

The labellers agreed only about 73% of the time, but the model learns one “best” answer.

Testing a model (ours)



five readings → all kept

All five readings stay, so we can see which ones a model covers and which it misses.

Where it helps Back to psychology

The differences show what a model is missing, which tells us what to train next.
Check how much raters agree, and also how they differ.

Averaging people hides their differences. A good test should keep them.

L. Ouyang *et al.*, *NeurIPS* 2022. **S. Hu***, X. Li* *et al.*, FIOVA, arXiv:2410.15270. *Equal contribution.

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Part 2 · A group

What Part 2 found

Whose standard does AI follow?

Part 1 · One person



Right answer, right reason?

Not yet: close scores, different mistakes.

Part 2 · A group



Whose standard does it follow?

A group of people, not one person.

Part 3 · A society



Like us enough to study us?

Part 4 · Your research



How can you use it?

So we should compare AI with a group of people, not just one person.

Next → So far, people have judged AI. Can AI play people?

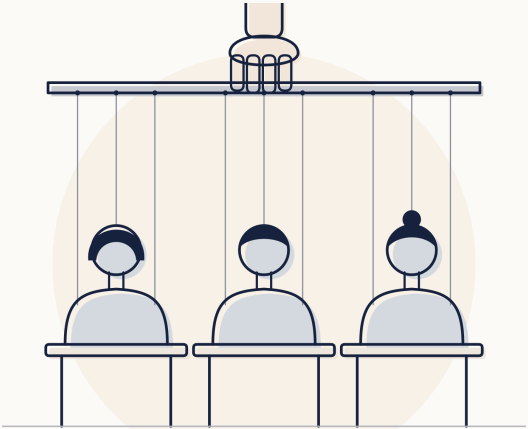
The route of this talk (P07), with Part 2's answer.

3

Part 3 · A society

AI can play people. Is it like us enough to study us?

Language models in the participants' place



Where Smallville came from

It began with a design problem: how do you get thousands of test users?

The designer's problem

A new online community is first tried out by a small group,



but trolls and fights show up only when thousands join.



Social Simulacra: a design in, users out

UIST writers' forum — generated by a language model

Pinned rule: **critique the writing, not the person**

Maya Smith a new Ph.D. student
"… I'm feeling really stuck."

Heather Hernandez an HCI professor
"It's normal to feel stuck …"

a troll
"You're just not cut out for this …"

far fewer posts like this

Language models played the users, troublemakers included. Different personas gave varied users, but each lived one moment.

many test users, 2022 —+ memory→ agents who live a day, 2023 —+ real people→ agents built from real people, 2024

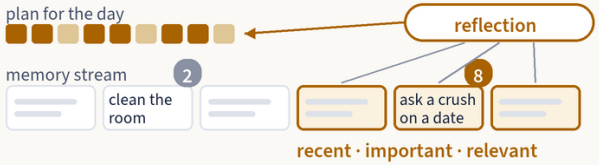
Smallville started from a simple need: lots of different test users.

J. S. Park *et al.*, "Social simulacra," *UIST* 2022 (Sec. 1 example). Drawings ours.

Smallville: a town of language-model agents

The next step was a town. What makes an agent seem like a person?





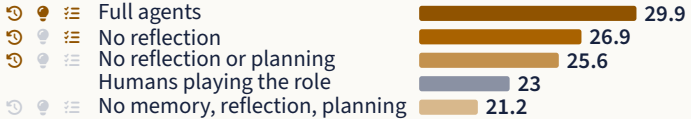
What it sees goes into a memory stream; it recalls what is recent, important and relevant, reflects on it, and plans its day.

How believable?

memory · reflection · planning

Take parts away, the scores fall.

100 people rated them (axis from 18)



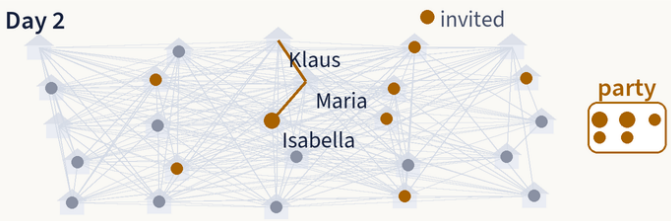
Memory, reflection and planning make agents seem real. But seeming real is not being real.

J. S. Park *et al.*, *UIST* 2023, Best Paper (Fig. 1). Citations: Google Scholar, 4 Nov 2026. Right: our drawings.

One party, a whole town

Seeming real is one thing. What happens when one agent plans a party?

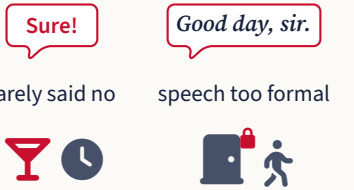
What spread, with nothing else scripted



12 → 5 invited, came

A foundation: group behaviour emerged.

Its limits, for now

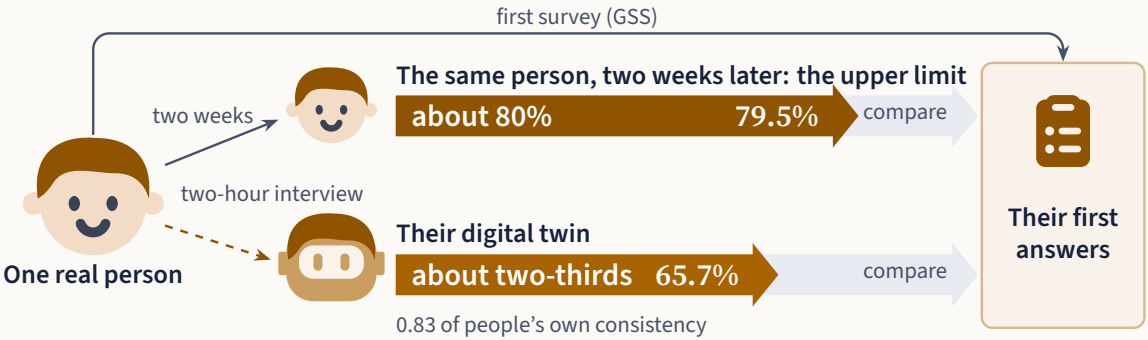


"should never be a substitute for real human input"

A big step, but we still don't know if the town behaves like a real one.

From 25 agents to 1,052 people

To find out, compare with real people. Can an agent stand in for one person?



1,052 people, one twin for each

To test a copy of a person, compare it with that same person, asked twice.

J. S. Park *et al.*, arXiv:2411.10109. Drawing: ours.

Our virtual students, with people as the reference

We check our virtual students against real people, in two directions.

The personalities here are examples we set for research. They don't describe any real student.

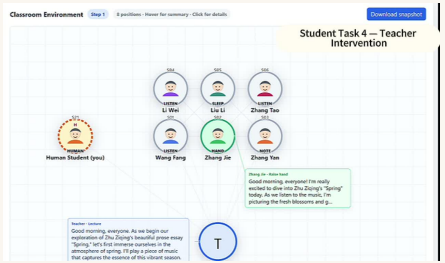
One person



Like a real student?

over time: one reply, then a whole session

Into a group





Like a real class?

across a class: one to many

Two tests against real people: over time, and across a class.

How we build a virtual student

First step: how do you build a more real agent, here a virtual student?

API route a closed model, a persona prompt: **the persona drifts back to a helpful expert**

Our route **an open-source model, fine-tuned on relevant materials**



We start from real lessons, teachers correct the replies, and we check every step.

Next → First over time, then across a class.

K. Li *et al.*, *COLM* 2024 (persona drift). Y. Ma*, **S. Hu*** *et al.*, EduEval, arXiv:2410.15701 (Sec. 3–4, App. C; LoRA on InternVL). EduPersona, EduVerse: arXiv:2510.04648, 2510.05650.

Would teachers notice?

Over time, start with one turn. Would teachers think these replies are real?

Two virtual students answer



"Who can tell us something about Mr Lu Xun?"

set outgoing: confident



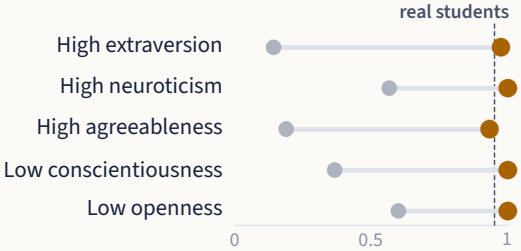
"Of course, I know Lu Xun is the founder of modern Chinese literature …"

set anxious: hesitates



"Well, uh, Mr Lu Xun, he's, uh, a, uh, very famous writer in China."

Share of replies judged real (0 to 1)



before training · after training · ten blind raters

After training: about as real as real students.

A reply can look real. That doesn't mean there's a mind behind it.

One student, a whole session

One turn is not enough: drift shows up over time. Do ours stay themselves for a whole lesson?

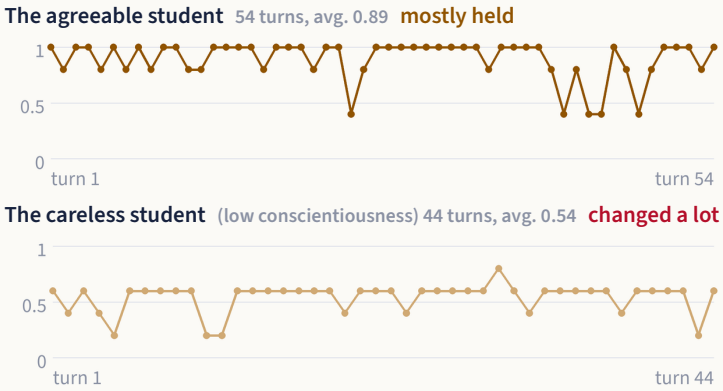
The test

Ten trainee teachers each taught five virtual students, one per personality: 50 sessions, 1,288 turns.

After every turn, a model ranks the five personalities.

1: the set one comes first;
0: not in the ranking.

Two of the longest sessions, redrawn from the paper.

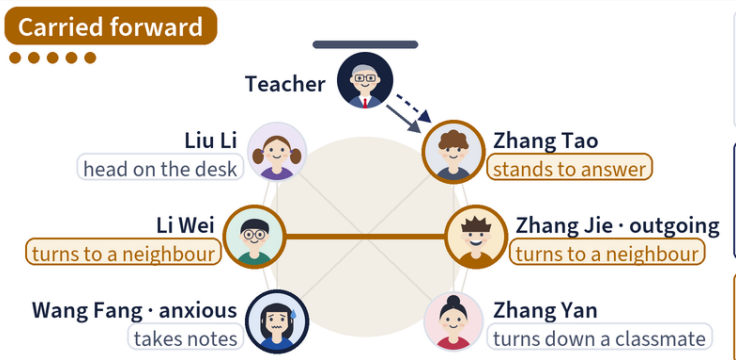


The persona mostly held, but one type changed a lot. Real students must judge it.

Y. Ma*, **S. Hu*** *et al.*, EduEval, arXiv:2410.15701 (Fig. 7, Table 7; redrawn). *Equal contribution.

Inside our simulated class

Now across a class: from one agent to many. Here is how the class is set up.



The teacher asks

"…peach, apricot and pear trees, all racing to bloom." Zhang Tao, how does this lively spring feel to you?

The teacher's feedback

"Zhang Tao, great to take part; keep to the question and you will understand more."

Zhang Tao's plan, carried forward

"Next time the teacher asks, I'll put my hand up, to stay focused."

In a class, each student reacts to the teacher, the room and the classmates nearby.

Y. Ma*, **S. Hu*** *et al.*, EduVerse, arXiv:2510.05650. Our drawing of one logged step (round table, step 6; translated). *Equal contribution.

shiyu.hu@ntu.edu.sg

Two students, one lesson

Does each student keep a persona among the others? Follow two of them to see.

One simulated lesson, step by step



Kept in a class

Each keeps its own pattern:

Zhang Jie: active, positive

Wang Fang: quiet, negative

Not always as we set

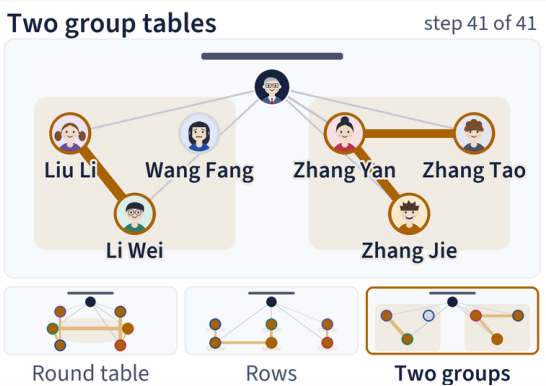
Her profile says she gradually joins in; in class she rarely does.

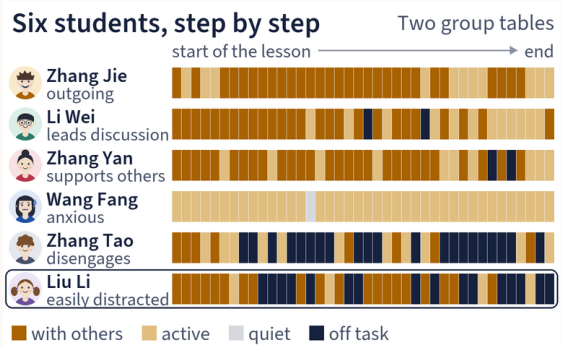
A neutral result: varied and in character, not always as designed.

In a class, each keeps a distinct persona, but not always the way we set it.

A classroom you can rerun

What a simulation is good for: change the room, run it again, watch the class unfold.





Nine runs: three seatings, three lesson texts.

A simulated class lets us explore how a class might unfold, within its limits.

Y. Ma*, **S. Hu*** *et al.*, EduVerse, arXiv:2510.05650. Drawn from our simulation logs; profiles from its appendix. *Equal contribution.

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What Part 3 found

Is AI like us enough to study us?

Part 1 · One person



Right answer, right reason?

Not yet: close scores, different mistakes.

Part 2 · A group



Whose standard does it follow?

A group of people, not one person.

Part 3 · A society



Like us enough to study us?

To rehearse a study, not to replace its people.

Part 4 · Your research



How can you use it?

To check a simulation, compare it with real people.

Next → One more question: how can you use this in your own study?

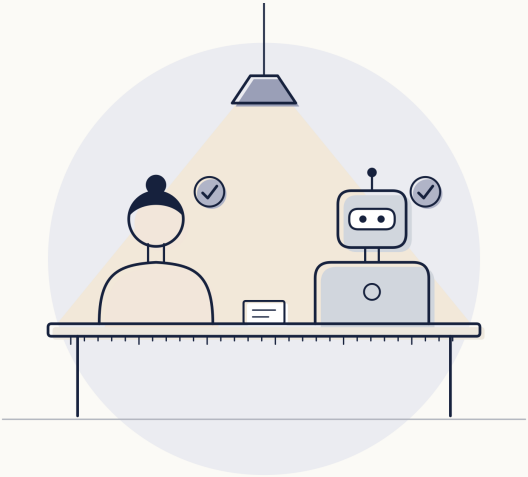
The route of this talk (P07), with Part 3's answer.

4

Part 4 · Your research

How can you use AI in your own study?

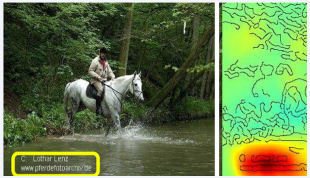
Two ways: analyse your data, model people



Multimodal learning: what did it use?

From Part 1: a right answer can rest on the wrong signal. When a model reads your data:

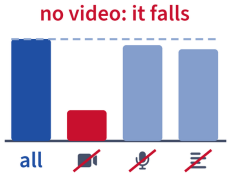
From Part 1: the horse photo



What did it use?

It said “horse”, but it read the tag. Video, voice, text: which one did yours use?

From my work in multimodal learning



Take one signal away

Drop video, voice or text, one at a time, and test again: what does the answer rest on?



Test where data changes

New people, new rooms, harder clips. In our tracking work: lost after a jump.

Take one signal away, and test where the data changes.

Evaluation: who is the reference?

From Part 2: people disagree. When you judge a model:

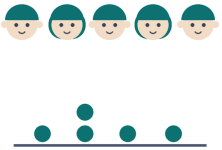
From Part 2: five people, one fall



Who is the reference?

A model describes, codes or rates your data. Against whom do you score it?

From my work in evaluation



A group as the reference

Compare the model with all your raters, not one answer or their average.



Look past the average

Same score, different mistakes: compare them item by item.

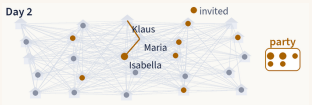
Compare with a group of people, and look past the average score.

From Part 2: **S. Hu***, X. Li* *et al.*, FIOVA (frame #640). Animations ours.

Agent modelling: does it stay itself?

From Part 3: an agent can seem real and still drift. When a model plays a person:

From Part 3: Smallville, our students



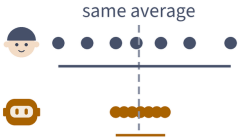
A town of agents: it seems real



One lesson: the persona still drifts

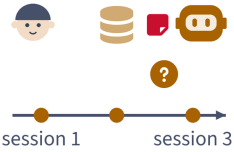
Your participant, or a partner who talks with them.

From my work in agent modelling



See the whole range

Compare with real people: the range of answers, not only the average.



Check it over time

Tell it something in session one, ask in session three. Fix what it decides; let the words vary.

Check it against real people: the whole range, and over time.

Closing

The whole talk on one page

What did each part tell us?

Part 1 · One person



Right answer, right reason?

Not yet: close scores, different mistakes.

Part 2 · A group



Whose standard does it follow?

A group of people, not one person.

Part 3 · A society



Like us enough to study us?

To rehearse a study, not to replace its people.

Part 4 · Your research



How can you use it?

Check it first, then use it.

Behavioral analysis — psychology helps AI

Role simulation — AI helps psychology

Studying the Human Mind with AI

AI learns from people. If we check it against people, it can help us study people.

The route of this talk (P07), answered part by part; the papers are cited on each part's pages.

Why psychology and the humanities matter more now

Two things I see, and how I would do research now

Two stories from maths

Jul 2026 · Claude
A counterexample to an 87-year-old conjecture

Oct 2026 · OpenAI
Claims results on hundreds of open problems
three papers withdrawn the next day

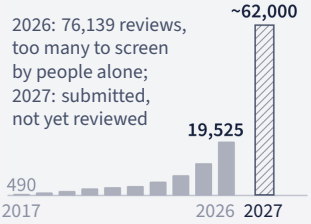
→ **PhD level, even expert**



Yann LeCun · Turing Award
"never going to get to human-level intelligence by just training on text"

ICLR submissions per year

2026: 76,139 reviews, too many to screen by people alone; 2027: submitted, not yet reviewed



→ **Making is cheap; judging is hard.**

How I would do research now

1 Outlast the model
Ask: would it still matter if models were ten times better?

2 Let AI do; you judge
Hand it the doing, keep the judging; write the first draft yourself.

3 Find your own crossing
Start from a question your field asks best, and bring AI to it.

When making things gets cheap, judging is the real work, and that has always been the job of psychology and the humanities.

T. Tao, blog, 21 Jul 2026; github.com/openai/math, 2026; W. T. Gowers, blog, 8 May 2026. OpenReview via Paper Copilot; ICLR Blog, 2026. Y. LeCun in G. Snyder, *Newsweek*, 2025; photo: École polytechnique, CC BY-SA 2.0.

Thank you

Questions and discussion welcome

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AI learns from people;
checked against people,
it can help us study people.

With thanks to Wenzhou-Kean University for the invitation.

Everything from this talk, in one place

Guides to each part of the talk, and the studies to read first. No code needed.

github.com/huuuuusy/studying-the-mind-with-ai

Start here	the talk in five minutes
Parts 1–4	key readings, and how they connect
Key readings	22 one-page cards, about 300 references
Glossary	35 terms in plain language
Reading path	step by step, up to the debates
Handout (PDF)	on paper, if GitHub is slow